Wouter Buijsman
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Second-order free probability

FREE PROBABILITY (ARXIV:2303.00713)

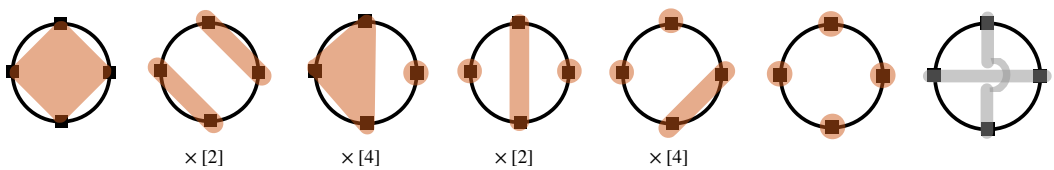
Object of interest: normalized trace

$$\phi(X) = \lim_{N\to\infty} \frac{1}{N}\mathbb{E}[\mathrm{Tr}(X)]. \tag{1}$$

Random matrix A: traces of powers can be expressed in terms of cumulants $k_n(A)$ using *free probability theory*:

$$\phi(A^n) = \sum_{\pi\in\mathrm{NC}(n)} k_\pi(A), \qquad k_\pi(A) = \prod_{b\in\pi} k_{|b|}(A), \tag{2}$$

where π is a partition with blocks b of size $|b|$. Example: the parition $\pi = (1,4)(2)(3)$ has block $(1,4)$ of size 2 and blocks (2), (3) of size 1. $\mathrm{NC}(n)$ gives the set of non-crossing partitions (illustrated below diagrammatically for $n = 4$):



First few powers:

$$\phi(A) = k_1(A), \tag{3}$$
$$\phi(A^2) = k_2(A) + k_1(A)^2, \tag{4}$$
$$\phi(A^3) = k_3(A) + 3k_2(A)\,k_1(A) + k_1(A)^3, \tag{5}$$
$$\phi(A^4) = k_4(A) + 2k_2(A)^2 + 2k_2(A)\,k_1(A)^2 + 4k_2(A)\,k_1(A)^2 + 4k_3(A)\,k_1(A) + k_1(A)^4. \tag{6}$$

The last line follows from the diagrams (it seems to contain a typo in the paper).

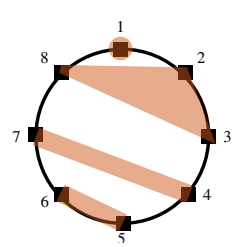
The cumulants of the distribution of $\mathrm{Tr}(A)$ can be expressed in terms of expectation values of $\mathrm{Tr}(A^n)$ by solving for $k_n(A)$, called "inversion" in the paper (k_4 omitted for brevity):

$$k_1(A) = \phi_1(A), \tag{7}$$
$$k_2(A) = \phi(A^2) - \phi(A)^2, \tag{8}$$
$$k_3(A) = \phi(A^3) - 3\phi(A^2)\,\phi(A) + 2\phi(A)^3. \tag{9}$$

A related diagrammatic approach can be applied to products of random matrices $A^{(1)}, A^{(2)}, \ldots, A^{(n)}$:

$$\phi(A^{(1)}\cdots A^{(n)}) = \sum_{\pi\in\mathrm{NC}(n)} k_\pi(A^{(1)}\cdots A^{(n)}), \qquad k_\pi(A^{(1)}\cdots A^{(n)}) = \prod_{b\in\pi} k_{|b|}(A^{(b(1))}\cdots A^{(b(n))}). \tag{10}$$

Example: the non-crossing partition π given by the diagram



gives a contribution

$$k_\pi(A^{(1)}\cdots A^{(8)}) = k_1(A^{(1)})\, k_3(A^{(2)}A^{(3)}A^{(8)})\, k_2(A^{(4)}A^{(7)})\, k_2(A^{(5)}A^{(6)}). \tag{11}$$

Cumulants $k_1(A^{(1)})$, $k_2(A^{(1)}A^{(2)})$ and so on can similar to the above be obtained through inversion, giving

$$k_1(A^{(1)} = \phi(A^{(1)}), \tag{12}$$

$$k_2(A^{(1)}A^{(2)}) = \phi(A^{(1)}\phi^{(2)}) - \phi(A^{(1)})\,\phi(A^{(2)}), \tag{13}$$

$$k_3(A^{(1)}A^{(2)}A^{(3)}) = \phi(A^{(1)}A^{(2)}A^{(3)}) - \phi(A^{(1)}\phi^{(2)})\,\phi(A^{(3)}) - \phi(A^{(1)}A^{(3)})\,\phi(A^{(3)}) - \phi(A^{(2)}A^{(3)})\,\phi(A^{(1)}) + 2\phi(A^{(1)})\,\phi(A^{(2)})\,\phi(A^{(3)}) \tag{14}$$

The third term of k_3 seemingly contains a typo as $A^{(2)}$ does not appear but $A^{(3)}$ appears twice. Main application (condensed matter physics / quantum information): higher-order eigenstate thermalization hypothesis.

SECOND-ORDER FREE PROBABILITY

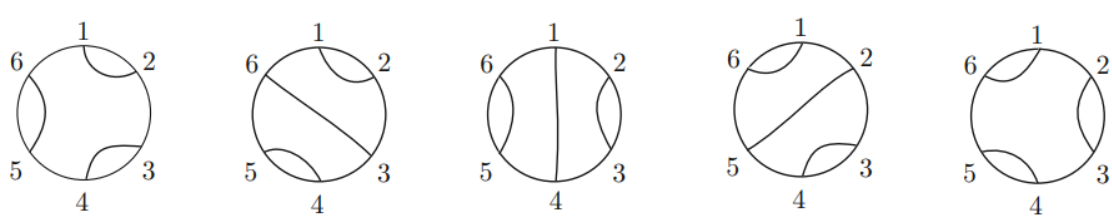
Reference: James A. Mingo and Roland Speicher, *Free Probability and Random Matrices*, (Springer 2017), Chapter 5. DOI: `10.1007/978-1-4939-6942-5`.

Single matrices

In order to introduce some notation: first the evaluation of traces of powers for normalized (!) matrices,

$$\alpha_k = \lim_{N\to\infty} \mathbb{E}[\mathrm{Tr}(X^k)]. \tag{15}$$

GUE: number of non-crossing pairings. Example: $\alpha_6 = 5$ with diagrams

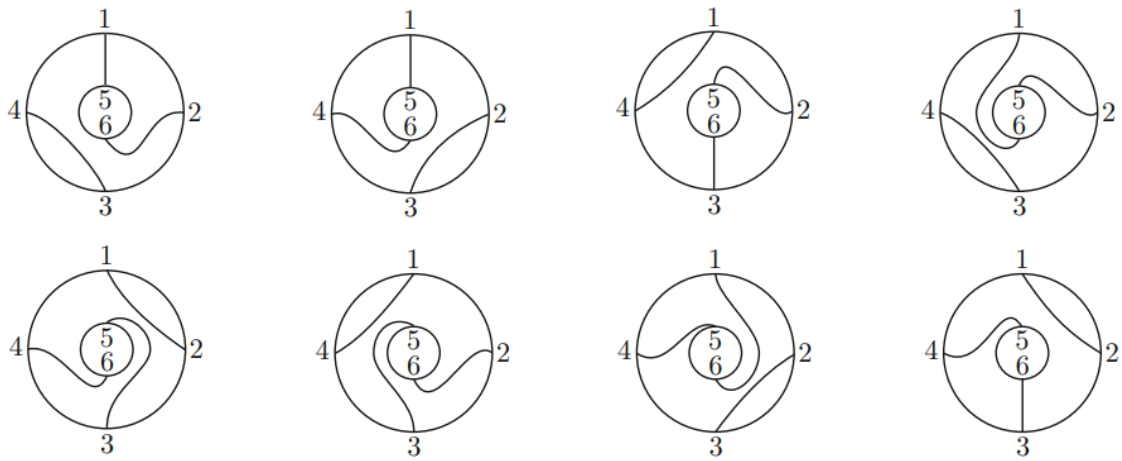


Covariance (beyond first-order free probability):

$$\alpha_{p,q} = \lim_{N\to\infty} \mathrm{cov}(\mathrm{Tr}(X^p), \mathrm{Tr}(X^q)) \tag{16}$$

GUE: number of non-crossing annular pairings. Non-crossing annular pairing: a pairing of the (p,q)-annulus is non-crossing if when we arrange the numbers $1, 2, \ldots, p$ in clockwise order on the outer circle and the numbers $p+1, p+2, \ldots, p+q$ counterclockwise order on the inner circle there is a way to draw the pairings so that the lines do not cross and there is at least one string that connects the two circles.

Example: $\alpha_{4,2} = 8$, with diagrams



Several matrices

Consider GUE's $A^{(1)}, \ldots, A^{(s)}$ and labels $1 \leq r_1, \ldots, r_{p+q} \leq s$. That is, each matrix is assigned a label ("colour"). Then

$$\lim_{N\to\infty} \operatorname{cov}(\operatorname{Tr}(X^{(r_1)} \ldots X^{(r_p)}), \operatorname{Tr}(X^{(r_{p+1})} \ldots X^{(r_{p+q})})) = \#\big(\mathrm{NC}_2^{(r)}(p,q)\big), \tag{17}$$

where $\mathrm{NC}_2^{(r)}$ denote the non-crossing annular pairings on the (p,q)-annulus with respect to the colour (π only connects matrices of the same colour).

Trivial example: all r_i's different gives zero (all different colours). For the covariance to be nonzero, the traces need to have at least one element in common.

Second-order cumulants, single matrices

Plan: develop a formalism as for first-order cumulants for the second order. For single matrices, define

$$\phi^2(A^p, A^q) = \alpha_{p,q}. \tag{18}$$

Each term gives the cumulant of a crossing:

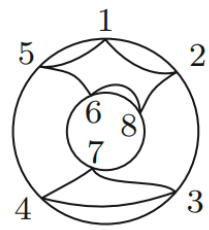
$$\phi^2(A^p, A^q) = \sum_{\pi \in \mathrm{NC}_2(p,q)} 1 \tag{19}$$

$$= \sum_{\pi \in S_{\mathrm{NC}}(p,q)} k_\pi \tag{20}$$

because only contribution 1 for a pairing contribution. A permutation π in $S_{\mathrm{NC}}(p,q)$ is a non-crossing permutation on the (p,q)-annulus if we can draw the cycles of π between the circles of the annulus so that:

- the cycles do not cross,
- each cycle encloses a region between the circles homeomorphic to the disc, and
- at least one cycle connects the two circles.

Example: $p = 5$ and $q = 3$ with $\pi = (1, 2, 8, 6, 5)(3, 4, 7)$ gives the diagram



Second-order cumulants, several matrices

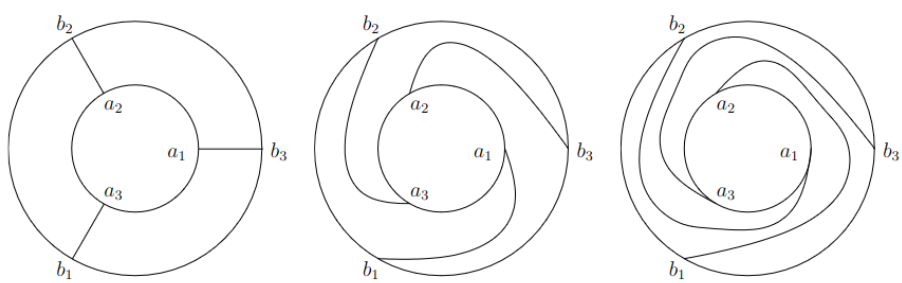
Warning: written in a hurry, might contain some typos.

One can show that

$$\phi^2(A^{(1)} \cdots A^{(p)}, B^{(1)} \cdots B^{(q)}) = \delta_{pq} \sum_{\pi \in \mathrm{SP}(n)} \phi_\pi(A^{(1)} \cdots A^{(p)}, B^{(1)} \cdots B^{(q)}) \tag{21}$$

where $\mathrm{SP}(n)$ denotes the set of all spoke diagrams and the A's on the one hand and B's on the other are free with respect to each other.

A non-crossing permutation of an (p, q)-annulus is called a spoke diagram if all cycles have just two elements (i, j) and the elements are on different circles, i.e.$i \in [p]$ and $j \in [q]$. We can only have a spoke diagram if $p = q$.



Example:

$$\phi^2(A^{(1)}A^{(1)}, B^{(1)}B^{(2)}) = \phi(A^{(1)}B^{(1)})\,\phi(A^{(2)}B^{(2)}) + \phi(A^{(1)}B^{(2)})\,\phi(A^{(2)}B^{(1)}) \tag{22}$$

As the A, B are not neccessarily free with respect to each other, rewrite this (Gabriel discussed the diagrammatics):

$$\begin{aligned}\phi^2(A^{(1)}A^{(1)}, B^{(1)}B^{(2)}) = {} & \phi(A^{(1)}A^{(2)})\,\phi(B^{(1)}B^{(2)}) - \phi(A^{(1)}A^{(2)})\,\phi(B^{(1)})\,\phi(B^{(2)}) \\ & - \phi(A^{(1)})\phi(A^{(2)})\,\phi(B^{(1)}B^{(2)}) + \phi(A^{(1)})\,\phi(A^{(2)})\,\phi(B^{(1)})\,\phi(B^{(2)}) \\ & + \phi^2(A^{(1)}A^{(2)})\,\phi(B^{(1)})\,\phi(B^{(2)}) + \phi(A^{(1)})\,\phi(A^{(2)})\phi^2(B^{(1)}B^{(2)}).\end{aligned} \tag{23}$$

Each term represents a cumulant:

$$\begin{aligned}\phi^2(A^{(1)}A^{(1)}, B^{(1)}B^{(2)}) = {} & k_2(A^{(1)}A^{(2)})\,k_2(B^{(1)}B^{(2)}) + k_2(A^{(1)}A^{(2)})\,k_1(B^{(1)})\,k_1(B^{(2)}) \\ & + k_1(A^{(1)})k_1(A^{(2)})\,k_2(B^{(1)}B^{(2)}) + k_1(A^{(1)})\,k_1(A^{(2)})\,k_1(B^{(1)})\,k_1(B^{(2)}) \\ & + k_{1,1}(A^{(1)}A^{(2)})\,k_1(B^{(1)})\,k_1(B^{(2)}) + k_1(A^{(1)})\,k_1(A^{(2)})k_{1,1}(B^{(1)}B^{(2)}).\end{aligned} \tag{24}$$

Here, second-order cumulants $k_{1,1}$ have been defined implicitly. Relations can be inverted to get expressions for these.